

Ensemble Methods: Boosting vs. Bagging

Final Project Report

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Abstract—This paper presents an empirical comparison of two ensemble learning paradigms—boosting and bagging—using decision-tree base learners. We implement a Decision Tree classifier, AdaBoost (with decision stumps), and a Random Forest from scratch. Controlled experiments on real-world datasets investigate scaling behaviour, noise robustness, and bias-variance decomposition. Unsupervised analysis (PCA, K-Means, DBSCAN) supplements the visual exploration of the data. Our findings show that *[TODO: summarise main result, e.g., “AdaBoost outperforms Random Forest on clean data but is more sensitive to label noise”]*.

Deliverables: In addition to this report, we provide the complete source code (tagged `v1.0-final`), a professional slide deck for the oral defense, and a separate contribution report detailing each team member’s specific work.

I. INTRODUCTION

The central question of this project is: “*Under what conditions does boosting outperform bagging, and vice versa, and why?*” To answer it, we implement the core algorithms from scratch and evaluate them on datasets of varying characteristics.

The remainder of the paper is structured as follows. Section II reviews essential literature. Section III describes our implementations. Section IV details the experimental setup and Section V presents the results. Section VI discusses the findings, and Section VII concludes. The accompanying slide deck and contribution report are submitted separately as required by the project brief.

II. RELATED WORK

A. Decision Trees

Decision trees recursively partition the feature space to minimise impurity [1]. CART [2] uses Gini impurity or entropy.

B. Boosting

AdaBoost [3] iteratively re-weights samples, giving higher importance to misclassified instances. The weight update rule is $\alpha_m = \ln \frac{1-\epsilon_m}{\epsilon_m}$ for binary classification.

C. Bagging and Random Forests

Bagging [4] reduces variance by training bootstrap replicates. Random Forests [5] add feature sub-sampling to decorrelate trees further.

D. Unsupervised Techniques

PCA [6] finds directions of maximum variance. K-Means [7] partitions data into k clusters, while DBSCAN [8] discovers clusters of arbitrary shape based on density.

III. METHODS

We implemented all algorithms in Python, using only NumPy (and `sklearn.metrics` for ARI). The code is available at <https://github.com/...> (tag `v1.0-final`).

A. Decision Tree

We follow the CART methodology with exhaustive split search. Both Gini impurity and entropy are supported, with `sample_weight` for the boosting module.

B. AdaBoost

The discrete SAMME variant [1] uses depth-1 stumps. Sample weights are updated each round and the stump is trained on weighted data.

C. Random Forest

Bootstrap samples are drawn with replacement. Trees are grown with feature sub-sampling (`sqrt` by default). OOB error is computed for each sample using trees where it was not in the bootstrap.

D. Unsupervised Pipeline

- **PCA:** Eigen-decomposition of the covariance matrix.
- **K-Means:** Lloyd’s algorithm with multiple restarts.
- **DBSCAN:** Brute-force region queries, expanding core points.

IV. EXPERIMENTAL SETUP

A. Datasets

We use three datasets (Table I).

TABLE I
DATASETS USED IN THE STUDY.

Dataset	Samples	Features
Breast Cancer Wisconsin	569	30
Adult Income	48 842	14
Covertypes (subset)	5 000	54

B. Preprocessing

Features were standardised (zero mean, unit variance). Missing values were dropped/imputed (justify). For the imbalanced Coverttype subset, we applied random oversampling of the minority class.

C. Metrics

We report accuracy, macro F1, and AUC-ROC (micro-averaged for multi-class). All experiments use 5-fold cross-validation with a fixed random seed (42).

D. Experiments

- 1) **Baseline:** Single tree vs. sklearn reference.
- 2) **AdaBoost scaling:** 1–200 stumps.
- 3) **Random Forest scaling:** varying trees and depth.
- 4) **Head-to-head:** 100 estimators, 5-fold CV.
- 5) **Noise robustness:** 5%, 10%, 20% label noise.
- 6) **Bias-variance decomposition:** 100 bootstrap replicates.
- 7) **Unsupervised analysis:** PCA scree, elbow, k-distance plots.

V. RESULTS

figures/example_scaling.pdf

Fig. 1. Test accuracy vs. number of estimators. (Placeholder)

Table II summarises the head-to-head comparison.

TABLE II
5-FOLD CV RESULTS (MEAN $\pm$ STD). (PLACEHOLDER)

Model	Accuracy	F1	AUC
Single Tree	0.85 $\pm$ 0.02	0.84 $\pm$ 0.03	0.91 $\pm$ 0.02
AdaBoost	0.92 $\pm$ 0.01	0.91 $\pm$ 0.02	0.96 $\pm$ 0.01
Random Forest	0.93 $\pm$ 0.01	0.92 $\pm$ 0.01	0.97 $\pm$ 0.01

The noise robustness experiment (Fig. 2) shows *[TODO: describe observation]*. The bias-variance decomposition (Table III) confirms that boosting reduces bias while bagging reduces variance.

figures/example_noise.pdf

Fig. 2. Accuracy degradation under label noise. (Placeholder)

TABLE III
BIAS-VARIANCE DECOMPOSITION ON BREAST CANCER DATASET.
(PLACEHOLDER)

Model	Bias ²	Variance
AdaBoost	0.05	0.10
Random Forest	0.08	0.06

VI. DISCUSSION

[TODO: interpret results, answer the central question, discuss when boosting outperforms bagging and why. Relate to dataset characteristics (noise, dimensionality, class imbalance). Comment on unsupervised findings: do cluster structures correlate with classifier performance?]

VII. CONCLUSION

[TODO: summarise key findings, mention limitations, and suggest future work (e.g., Gradient Boosting, t-SNE).]

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